

Digital Signal Processing - Take-home Assignment

Assignment Overview

In this assignment, you will implement the two “pillars” of DSP: Convolution and the DFT. You are expected to write your own functions rather than relying solely on high-level library shortcuts like `numpy.convolve` or `numpy.fft`.

Part 1: Convolution (The Input Side Algorithm)

Convolution is “the process of combining two signals to form a third signal.” Specifically, it is used in the **Input Side Algorithm**, where each input sample is replaced by a scaled version of the impulse response.

Task:

1. Generate a “noisy” input signal consisting of a 50 Hz sine wave sampled at 1000 Hz for 1 second.
 2. Define an impulse response for a **3-point Moving Average Filter**.
 3. Write a Python function `manual_convolution(x, h)` that implements the convolution using the **Input Side Algorithm** (nested loops).
 4. **Compare:** Use `matplotlib` to plot the original signal and the filtered signal.
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Part 2: The Discrete Fourier Transform (DFT)

The DFT is the mathematical tool used to find the frequency spectrum of a discrete signal. While the FFT (Fast Fourier Transform) is the industry standard for speed, understanding the DFT “correlation” method is vital.

Task:

1. Create a signal that is the sum of two sine waves (e.g., 60 Hz and 120 Hz) plus random white noise.
2. Implement a function `calculate_dft(x)` that calculates the Real and Imaginary parts of the frequency domain using the equations:

$$\text{Re}X[k] = \sum_{i=0}^{N-1} x[i] \cos(2\pi ki/N)$$

$$\text{Im}X[k] = - \sum_{i=0}^{N-1} x[i] \sin(2\pi ki/N)$$

3. Calculate the **Magnitude** of the frequency domain:

$$\text{Mag}X[k] = \sqrt{(\text{Re}X[k])^2 + (\text{Im}X[k])^2}$$

4. Plot the Magnitude Spectrum. Label your axes correctly (Frequency in Hz vs. Amplitude).
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Submission Requirements

- **Code:** A single `.ipynb` (Jupyter Notebook) file.
- **Plots:** All plots must have titles, gridlines, and axis labels. They should appear inside your `.ipynb` notebook.